

100 Days of Azure DevOps

Day 54 Handout — Run one image, get a DNS label, delete the wrapper — no node pool required

6 - Containers & Kubernetes · Personal lab only · Educational content · Not a sales pitch

Architecture A — ACI vs AKS vs Container Apps (same image, three runtimes)

Dimension	ACI	AKS / Container Apps
What you operate	A container group. No kubelet, no node pool	AKS: control plane + nodes. Container Apps: scale rules, no cluster YAML
Bill	vCPU/memory/second while allocated. Easy to forget overnight	AKS: nodes + control plane even at idle. CA: consumption or dedicated
Pull from ACR	AcrPull on a managed identity (--acr-identity). Not admin password in history	kubelet / CA revision uses AcrPull or imagePullSecrets
Network	Public IP + --dns-name-label, or VNet-injected group	Ingress / internal load balancer / CA ingress
Fit	Job, demo URL, sidecar-less hello	AKS when you need kube API. CA when you want scale without kube YAML

Architecture B — decide fast-food vs gym membership

Constraint	Default today	Move off ACI when
Need a URL tonight	az container create, unique dns-name-label, port the app listens on	You need Deployments, HPA, or Helm — that is AKS, later
Need scale to zero / HTTP scale	ACI is not that product	Azure Container Apps. Do not fake scale-to-zero with ACI restart policy
Need kube API	Stay on ACI for the hello	Day 56 AKS — with a destroy date
Identity to ACR	User-assigned identity + AcrPull + --acr-identity	Registry username/password. Lab last resort, not the pattern
After the taste	az container delete the same session	Leaving ACI up 'to feel cloud-native' is a silent bill

Architecture C — pitfalls that surprise-bill or never start

Pitfall	What you see	Fix
dns-name-label taken	Create fails on the name, not on the image	Pick a unique label. Azure is not being rude
Wrong port	Group Running, browser hangs	containerPort / --ports must match the process (3000 vs 80)
Pull denied	Waiting / CannotPull / unauthorized	AcrPull on the identity; image tag exists; not Hub-by-accident
Admin creds in CLI history	Password in shell transcript	--assign-identity + --acr-identity
Forgot delete	Next week's Cost Analysis mystery	Delete is part of the lab. Restart policy Never for a one-shot

One-liner to remember

ACI is container fast-food — no cluster gym membership required. Taste the URL, throw away the wrapper. Fast-food left on the table still appears on the card.

Step-by-step lab (personal Azure / Azure DevOps only)

Personal subscription. rg-day54 (or reuse rg-day52). Delete the container group before you close the laptop.

- 1 Create a user-assigned identity in the lab RG. Grant it AcrPull on the Day 52 registry. Do not enable ACR admin to 'just run ACI'.
- 2 az container create with --image <acr>.azurecr.io/myapp:<Build.BuildId or day52>, --ports matching the process, --dns-name-label unique, --assign-identity and --acr-identity on that identity, --restart-policy Never, small CPU/memory.
- 3 Wait until ProvisioningState Succeeded. Hit http://<label>.<region>.azurecontainer.io (or the IP). If it fails, az container logs — do not immediately jump to AKS.
- 4 az container delete -g ... -n hello-aci --yes. Confirm the group is gone. ACI left running is the lab failure even if the URL worked.
- 5 Write docs/aci-day54.md: image tag used, whether identity pull worked, delete receipt. Optional: note when you would pick Container Apps instead (HTTP scale) vs AKS (kube API).
- 6 Do not keep ACI overnight. Do not create AKS today to feel advanced.

Starter CLI — pull with AcrPull, then delete

```
az identity create -g rg-day54 -n id-aci-pull
ACR_ID=$(az acr show -n <uniqueacr> --query id -o tsv)
PID=$(az identity show -g rg-day54 -n id-aci-pull --query principalId -o tsv)
az role assignment create --assignee $PID --role AcrPull --scope $ACR_ID
ID=$(az identity show -g rg-day54 -n id-aci-pull --query id -o tsv)
az container create -g rg-day54 -n hello-aci \
  --image <uniqueacr>.azurecr.io/myapp:day52 --os-type Linux --cpu 1 --memory 1 \
  --ports 3000 --dns-name-label <unique> --restart-policy Never \
  --assign-identity $ID --acr-identity $ID
az container logs -g rg-day54 -n hello-aci
az container delete -g rg-day54 -n hello-aci --yes
```

Done checklist

- One successful ACI run of the ACR image using AcrPull (not Hub)
- DNS/port actually answered, or logs explain why — then fixed
- Container group deleted in the same session
- docs/aci-day54.md notes identity vs admin-password temptation

Tomorrow — Day 55

Kubernetes fundamentals — airport vocabulary without buying the airport yet